# E · TOTEM

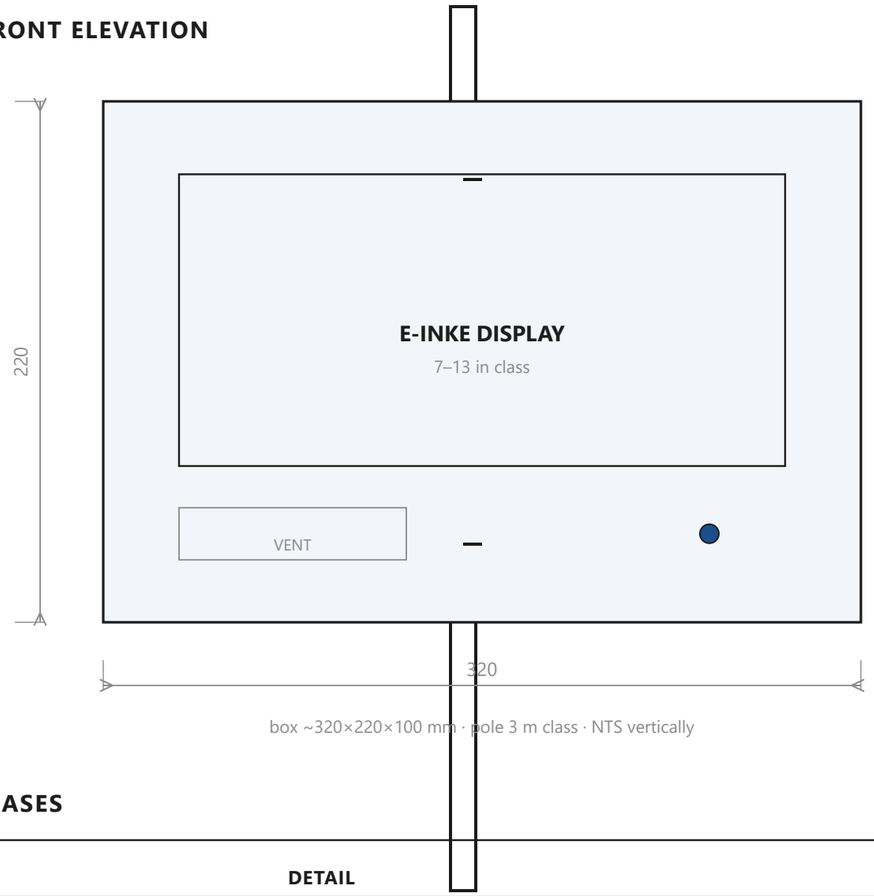
GENERAL ARRANGEMENT — COMMUNITY BOARD

**AF-E-101**

REV A · 2026-11-12

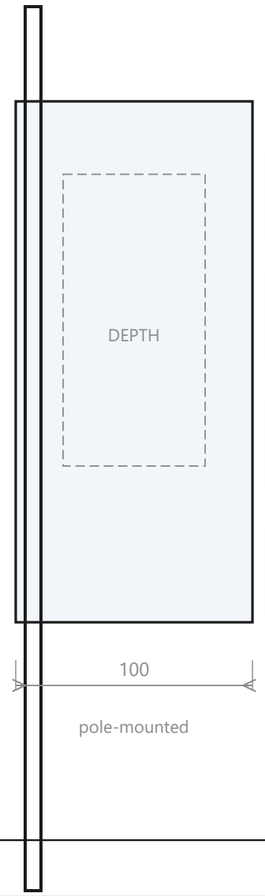
**FOR REVIEW**

## FRONT ELEVATION




box ~320×220×100 mm · pole 3 m class · NTS vertically

## SIDE ELEVATION




pole-mounted

## CONTENT PIPELINE

**PUBLISHER**
community / agency signs content

↓

**MESH**
signed envelopes flood the network

↓

**TOTEM**
verify → render → E-Ink

↓

**PUBLIC**
sunlight-readable · zero idle power

## USE CASES

| USE | DETAIL |
|---|---|
| Emergency alerts | wildfire evacuation, weather — mesh-wide in minutes |
| Community notices | council, school, clinic, events — the digital general-store board |
| Agency bulletins | province, RCMP, health unit — signed content only |
| Local info | road conditions, water advisories, powwow/community events |
| Mesh status | optional node-health display — coverage made visible |

## WHY IT'S UNIQUE

— Only public noticeboard that IS a network node — it routes, repeats, stores.

— Signed content: every notice carries publisher identity — no anonymous flyers.

— Works when grid AND cell network are both down — the blackout case.

— EN/FR pipeline; community controls what appears — sovereignty in public space.

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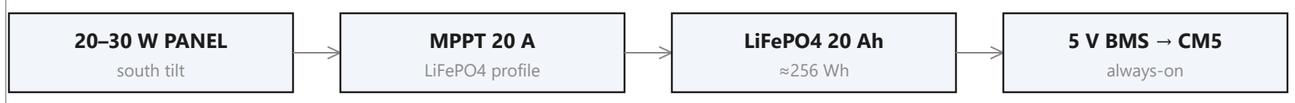
| | |
|---|---|
| DRAWN | **Surafel Negussie** |
| SHEET | **1/3** |
| DATE | **2026-11-12** |
| DWG NO | **AF-E-101** |
| SCALE | **NTS** |
| UNITS | **mm** |

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## PARTS LIST — PER TOTEM

| # | PART | SUPPLIER | APPROX CAD | QTY |
|---|---|---|---|---|
| 1 | CM5 4GB + IO Board (same as L1 — fleet common) | PiShop / Canada Computers | $90–130 | 1 |
| 2 | 13.3in or 7.5in e-Paper HAT (SPI) | Waveshare / DFRobot | $100–200 | 1 |
| 3 | SX1262 LoRa breakout 915 MHz | DFRobot / AliExpress | $15–25 | 1 |
| 4 | SIM7600 LTE HAT (optional backhaul) | DFrobot / Waveshare | $60–90 | 0–1 |
| 5 | 20–30 W poly solar panel + MPPT 20 A | Amazon.ca / Renogy | $120–200 | 1 |
| 6 | 12.8 V 20 Ah LiFePO4 battery | Amazon.ca / supplier | $180–260 | 1 |
| 7 | IP66 polycarbonate box ~320×220×100 | Bud Industries / Amazon.ca | $80–140 | 1 |
| 8 | Pole + U-bolts + window film + gasket | hardware supplier | $60–100 | 1 |

## POWER SYSTEM




Autonomy ≈3–5 days no-sun at 2–4 W avg; oversized panel is the winter plan.

## CONTENT PIPELINE — SIGNED PUBLICATIONS

— Publishers sign content (Ed25519) — community/agency keys.
— Mesh floods signed notice envelopes; node verifies before render.
— Rotation: alerts > notices > info; expiry timestamps auto-clear.
— Rate-limit + dedup; offline queue preserves order; admin can pin/hide.

## INSTALLATION — TWO PERSON, ~30 MIN

1. Set pole (post-hole or existing utility pole U-bolt).
2. Hang box on rear hinge; face display to road/foot traffic.
3. Connect solar leads; route antenna ports top, drip-loop cable.
4. Power on; verify mesh join + first content render.
5. Service: front latch → SD card + battery swap <10 min.

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| | |
|---|---|
| DRAWN | **Surafel Negussie** |
| SHEET | **2/3** |
| DATE | **2026-11-12** |
| DWG NO | **AF-E-102** |
| SCALE | **NTS** |
| UNITS | **mm** |

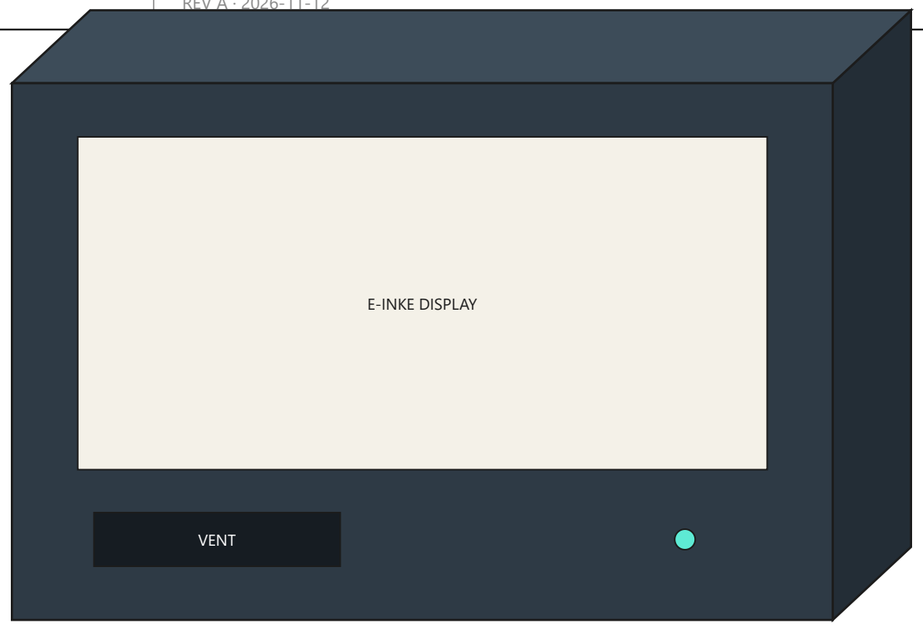
# E · TOTEM

FINISH & MATERIALS — FINAL LOOK

**AF-E-103**

REV A · 2026-11-12

**APPEARANCE REV A**




E TOTEM — FRONT

~320 × 220 × 100 mm box on 3 m pole · IP66

## COLOR / FINISH OPTIONS

**CHARCOAL**
default civic color

**FOREST**
park/natural areas

**COMMUNITY RED**
custom per community

## MATERIALS SPEC

| COMPONENT | MATERIAL | FINISH | NOTE |
|---|---|---|---|
| Enclosure | polycarbonate IP66 box | texture, UV stable | Bud Industries class |
| Display | E-Ink 13.3in + window film | matte anti-glare | sunlight readable |
| Pole | galvanized steel / treated wood | powder-coat charcoal | community install |
| Gasket | silicone continuous bead | — | IP66 "driving rain" |
| Labels | vinyl-cut bilingual | reflective | EN/FR community ID |

## DESIGN NOTES

— Charcoal powder-coat hides fingerprints and 10-year weathering.

— Community name + nation badge on lower bezel — vinyl, field-replaceable.

— Display window recessed 8 mm — survives thrown objects, hockey pucks.

— Totem IS the landmark — finish should read 'public infrastructure', not gadget.

— Solar panel separate, same charcoal frame — visual unity.

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DRAWN **Surafel Negussie**
SHEET **3/3**
DATE **2026-11-12**
DWG NO **AF-E-103**
SCALE **NTS**
UNITS **mm**